



The Effect of Words' Visual Complexity Distribution on Saccade Targeting in Reading Chinese Sentences

An Eye Movement Decision and Motor Control Experiment

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Abstract

Previous studies about the visual and linguistic factors that influence the decision about *where* to move the eyes next in reading suggest a strong influence from low-level features; some studies also assume that this decision involves the computation of a saccade target before the oculomotor program is executed. In order to test whether the distribution of visual components within Chinese words influence the saccade targeting mechanism, we devised an index that reflects the distribution of visual information along 2-character words' area called the Visual Complexity Distribution (VCD) index. Three groups of words with a marked VCD index (i.e. Left-Bias, Right-Bias and Non-Bias) were identified and embedded in natural sentences; the eye movement of Chinese native speakers was recorded while they read these sentences in order to contrast first-pass duration and landing position eye movement measures between conditions. The experimental effects were estimated through contrast between conditions using Linear Mixed Models. The analyses on initial fixation position indicated a rightwards shift when sending the eyes towards words with Right-Bias in comparison to the other two conditions and shorter fixation durations when biased words are fixated in comparison to the Non-Bias words. These results indicate that the luminance pattern within 2-character Chinese words, as reflected by the VCD index, can influence the specification of a saccade target when those words are about to be fixated as well as modulate the fovea load when those words are currently fixated.

Introduction

The act of reading can be apprehended by studying the eye movement pattern along the lines of a text; in a basic level, this pattern can be decomposed in two different ocular behaviors: *saccades* and *fixations*. **Saccades** are small and fast jumps towards different location in the page/screen, movements that bring new visual information to the fovea in order to be further processed by the visual cortex; alternatively, **fixations** are the halts between saccades and represent the time that the eyes remain in one position before the next saccade is executed. Thus a fixation duration indicate the ease or difficulty in word identification, lexical access and passage integration (Rayner, Pollatsek, Ashby, & Clifton Jr, 2012; Reichle, Pollatsek, Fisher, & Rayner, 1998). As reading proceeds, the lexical mapping of each word takes place and the overall meaning of the sentence is gradually enriched by the information acquired with each fixation; thus, the interplay between these oculomotor behaviors is informative about ongoing cognitive processes and constitutes the variables of interest in eye movement experiments.

Several studies have examined the influence of orthographic language features on oculomotor control and saccade programming in alphabetic languages, such as *word length* and *inter-word spacing* (Rayner, 1979; Rayner, Fischer, & Pollatsek, 1998; McConkie, Kerr, Reddix, & Zola, 1988). Rayner (1979) was the first author to propose the existence of a *saccade strategy*, making reference to the oculomotor mechanism responsible for deciding where to send the eyes next; this statement was inferred from the proportion of *Initial Landing Position* (ILP) of inter-word saccades in words of different lengths. Rayner found that the ILP within a word is modulated by its length and when making inter-word forward saccades the reader's eyes tended to land half way between the end and the middle of the word; that is to say, word length information is used by English readers for moving their eyes along the text and thus choosing saccade targets. This study proposed the existence of a *Preferred Viewing Location* (PVL) near the word center, result that was taken as evidence supporting a word-based saccade strategy in reading English, finding that was later replicated in reading other languages such as German, Hebrew and Uighur (Deutsch & Rayner, 1999; Nuthmann & Kliegl, 2013; Yan et al., 2014).

To estimate the relative weight of linguistic and visual features on the saccade targeting mechanism in reading different languages, first we must acknowledge the possibility that different writing systems provide specific cues that might draw the native readers' eyes with differential intensity. For example, the validation of the PVL phenomenon across alphabetic languages indicates that inter-word spaces can be regarded as salient visual cues readily available from parafoveal vision, information actively used in deciding where to send the eyes next. The present study attempts to address the question about the relative weight of the visual features of traditional Chinese characters upon the native Chinese readers' *saccade strategy* when reading natural sentences lacking of inter-word spaces; for doing it so and in view of the results of a flat PVL within 2-character words (Yang, 1997; Yang & George, 1999), we devised an index that reflects the *Visual Complexity Distribution* (VCD) within 2-character Chinese words. This index reflects the distribution of visual components along the words' area, which we consider to be of relative importance in the decision about *where* and *when* to move the eyes when reading Traditional Chinese.

Main Objectives

1. Examine what visual cues are used by native Chinese readers for oculomotor control and saccade programming while reading natural traditional Chinese sentences.
2. Introduce and validate an index for 2-character Chinese words that reflects the distribution of visual components along the words' area and create different and distinctive experimental conditions from it.
3. Test the VCD index's influence on first-pass duration and Initial Landing Position (ILP) eye movement measures between the identified experimental conditions.

Material

Frequently, the variable used to represent the characters' visual complexity has been *stroke count* either at the character or at the word level; however and as several findings suggest that saccades are sent towards word units and not towards individual characters neither towards sub-lexical units (Yang & George, 1999; Yan, Kliegl, Richter, Nuthmann, & Shu, 2010; Yan, Kliegl, Shu, Pan, & Zhou, 2010; Zang, Meng, Liang, Bai, & Yan, 2013), a new index that captures the *Visual Complexity Distribution* (VCD) along the words' area was devised and tested experimentally. We computed the VCD index of 2-character words from the *Academia Sinica Chinese Corpus* by displaying each word in a monitor screen using *kai* font with a size of 32^2 pixels in black color on white background; following a similar procedure for the computation of *stroke density* in Cai, Chi, and You (2001), we used a MATLAB

script that allowed us to retrieve the color values of each pixel, make a count of the pixels filled by the characters' strokes within the word divided horizontally into 8 bins and calculated the slope of the regression line along these bins as shown in Figure 1. The number of bins used to divide the words' area was found to be sensible enough as to reflect the variability of visual components along horizontally displayed words.

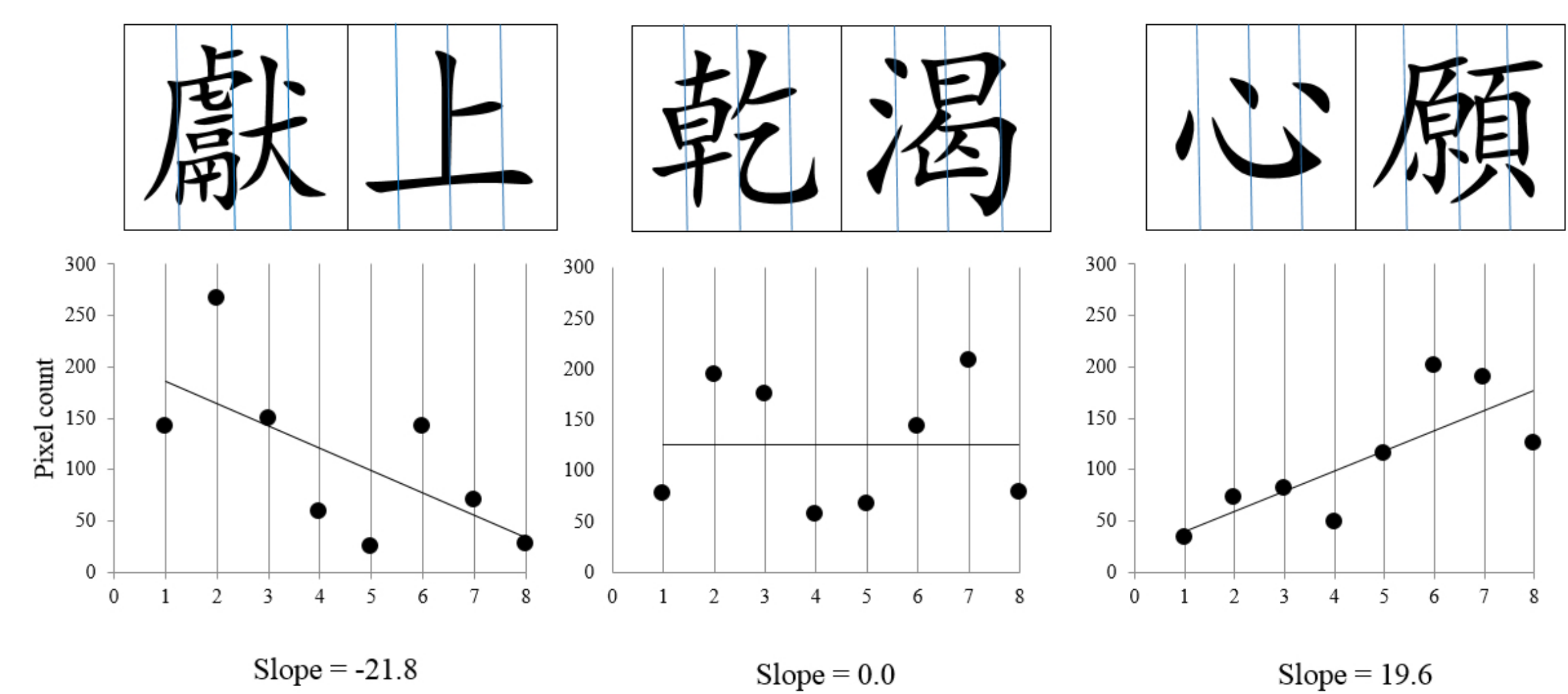


Figure 1: Visualization of the words' VCD Index computation

The VCD index's value and sign reflect the distribution of visual components or the luminance pattern within 2-character words, information which might be readily available from parafoveal vision and aid the selection of saccade targets. By using the VCD index we identified three groups of 42 target words each with similar *word frequency*, *neighborhood size* and *word stroke number*, and labeled them by their VCD bias: *a*) Negative VCD or Left Bias (LB) (i.e. $VCD \leq -10$), *b*) VCD index around 0 or Non Bias (NB) (i.e. VCD between -0.09 and 0.09) and *c*) Positive VCD or Right Bias (RB) (i.e. $VCD \geq 10$). The target words were embedded in natural sentences of 20 to 27 characters in length and always occupied a central position avoiding the first and the last five character positions; additionally and in order to reduce the context influence on the saccade targeting mechanism, the predictability of the target words was controlled and kept in a fairly low value for all of the target words within our reading material. Finally, the syntactic category of target words was counterbalanced within each condition, half of the target words were *verbs* and the other half were *nouns*.

Participants

Forty-two first-year psychology students from National Chengchi University (36 women and 6 men) participated in the experiment for partial fulfillment of course credits. All participants were Taiwanese, native Chinese speakers, had normal or corrected-to-normal vision and were naive concerning the purpose of the experiment. Their ages were between 19 and 21 years of age ($M = 19.7$; $SD = 0.7$).

Apparatus and Methods

A SR-Research EyeLink 1000 eye tracking system was used to monitor and record the participants' eye movement at a rate of 1000 Hz. Viewing was binocular but only the participant's dominant eye was recorded; participants read single-line Chinese sentences from a computer monitor (resolution: 1024×768 pixels), the characters were displayed in black color in light gray background and the reading direction of the sentence was horizontal from left to right. The characters were displayed in *kai* font type in a size of 32^2 pixels with a space of 4 pixels between each character; in this way, at a viewing distance of 70 cm, one character with the space before it subtended approximately 1° of visual angle. Additionally a chin rest was used to minimize the head movements and a response console with left and right keys served as the interface for answering the comprehension questions as well as to terminate the trials. First, participants read 8 practice sentences and 2 comprehension questions for getting to familiarize with the experimental procedure; after solving the doubts about the experiment, participants started to read the experimental sentences. Each participant read 138 experimental sentences in 3 blocks and the first 4 sentences of each block were fillers whose data was excluded from subsequent analysis. Sentences were presented once to each participant with the three conditions counterbalanced over participants following a Latin square design and the experiment took about 40 minutes to complete. After the eye movement data was collected, the fixations were determined with an algorithm for binocular saccade detection (Engbert & Kliegl, 2003).

Variables definition and Data Analyses

The eye movement measures that were thought to be relevant to our research question are: *a*) *First Fixation Duration* (FFD), it is the time of the first fixation on the target word irrespective of the number of fixation and is considered to be a measure that reflects a relatively early stage of lexical access and fovea processing; *b*) *Single Fixation Duration* (SFD), it is the fixation time on a target word that is fixated only once and it is a measure of early processing and lexical access; and *c*) *Gaze duration* (GD), it is the sum of all the first pass fixations on the target word before the eyes are sent to another word, measure that reflects the integration of visual and semantic features of the word. Specifically and in order to test the influence of the words' VCD index on saccade targeting, we were interested in the ILP within the selected target words and particularly on the ILP of single fixation cases. Thus the statistical analyses presented below are based on those fixations that followed forward first-pass saccades and involved the Region of Interest (ROI) containing the target words. The final number of eye movement data that remained after the data cleaning was 3727 for *first fixation cases*, 2996 for *single fixation cases* and 3618 for *gaze duration*. Statistical inferences about the VCD index's effects

on these eye movement measures are based on Linear Mixed Models (LMMs) for continuous duration and position variables. The experimental effects were estimated by a contrast matrix between pair of conditions; in the main model, the RB condition was the base line which was contrasted against both, LB and NB conditions (i.e. LB - RB and NB - RB). Similarly, the alternative model regarded NB target words as the base line and by doing it so, the contrast between experimental conditions was completed (i.e. LB - NB). As the groups of target words were found to be comparable in several linguistic and visual variables, the estimated effects provide reliable statistical inferences about the relative VCD influence for each of the eye movement variables under analysis. These methods allowed us to estimate the fixed effects for each condition and simultaneously consider crossed random effects from participants and sentences. Additionally and given the large influence that have been attributed to the *launch site* on ILP, we considered the launch site of each initial fixation as a covariate term or confounding factor; this allowed us to measure its main effect and partial it out in order to estimate the experimental effects on each of the aforementioned eye movement variables. All of the statistical procedures were done using the statistical software R, version 3.1.1 (Team et al., 2013), multivariate analyses were carried out using the *lmer* function of the *lme4* package (Bates, 2010) and the figures were created with the *ggplot* package for data visualization (Wickham, 2016). Finally, we used the *remef* package (Hohenstein & Kliegl, 2015) for removing the random factors variance and fixed effects from the dependent variable of each model.

Results

The descriptive statistics are graphically presented in Figure 2 for each of the eye movement variables mentioned above, in this figure each bar stands for a particular VCD condition and the vertical line on it denotes 1.5 times the standard error of the mean. Table 1 presents the fixed effects of the planned contrasts for two models along different eye movement measures; the *main model* (upper part), contrasted the responses on LB and NB target words against RB words; in the alternative model (bottom part) the observations on LB and RB target words are contrasted with those on NB target words, and only the effects for the contrast LB - NB are presented because the second contrast is common between both models.

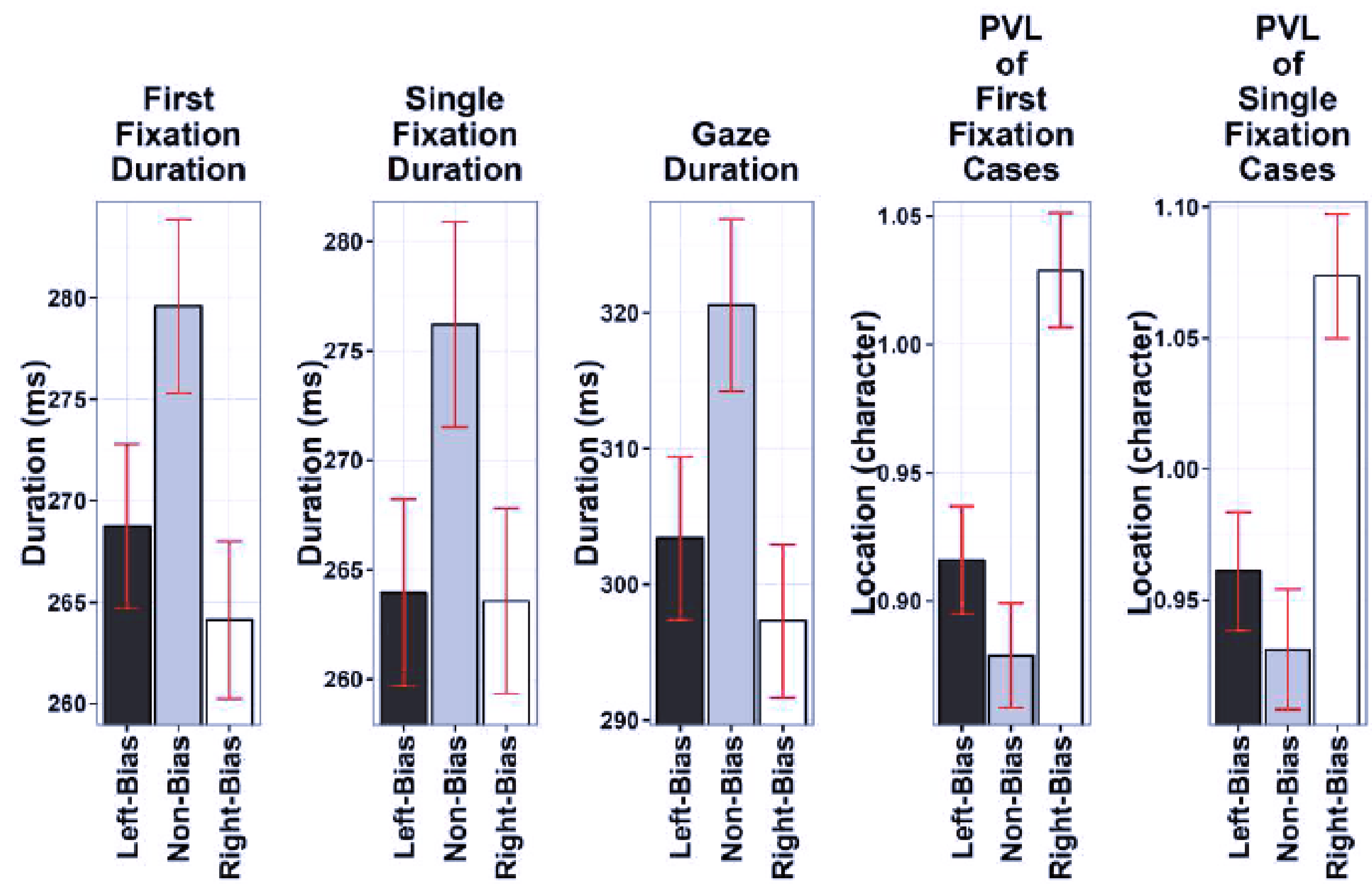


Figure 2: Mean for each eye movement measure by VCD condition

Figure 2 shows that the word identification or duration measures (i.e. FFD and SFD) as well as the total fixation time (i.e. GD), were significantly larger for the NB condition than for the LB and RB conditions; reaching significant levels for the contrast LB - NB (FFD: $t = -2.14$; SFD: $t = -2.41$; GD: $t = -2.32$; all $p < 0.05$) as well as for the contrast NB - RB (FFD: $t = 3.06$; SFD: $t = 2.52$; GD: $t = 3.00$; all $p < 0.05$). Instead, the contrast between LB - RB conditions didn't reach significant levels (FFD: $t = 0.92$; SFD: $t = 0.13$; GD: $t = 0.69$; all $p > 0.3$). Additionally and for the position measures it was found that the PVL of the RB condition was relatively further into the word in contrast to LB and NB conditions for both sets of initial fixations, reaching significant levels in the LB - RB contrast (PVL FF: $t = -4.81$; PVL SF: $t = -4.73$; all $p < 0.01$), as well as in the NB - RB contrast (PVL FF: $t = -6.40$; PVL SF: $t = -5.94$; all $p < 0.01$). Furthermore, the PVL of LB and NB conditions was found to be close from each other, as suggested by the lack of statistical significance in the contrast LB - NB (PVL FF: $t = 1.60$; PVL SF: $t = 1.25$; all $p > 0.1$).

Table 1: Estimated effects on eye movement measures using LMM

Model	Effects	FFD	SFD	GD	PVL FF	PVL SF
Main	LB – RB	0.02 (0.02)	0.002 (0.02)	0.02 (0.02)	-0.11 (0.02) **	-0.11 (0.02) **
	NB – RB	0.05 (0.02) **	0.05 (0.02) *	0.07 (0.02) **	-0.15 (0.02) **	-0.14 (0.02) **
	Launch Site	-0.004 (0.01)	0.01 (0.01)	-0.03 (0.01) **	0.36 (0.01) **	0.34 (0.01) **
	Intercept	5.56 (0.02) **	5.55 (0.02) **	5.66 (0.03) **	0.94 (0.04) **	0.99 (0.03) **
Alternative	LB – NB	-0.04 (0.02) *	-0.04 (0.02) *	-0.05 (0.02) *	0.04 (0.02)	0.03 (0.02)
	Observations	3573	2851	3464	3573	2851

The standard errors of estimated effects are reported in parenthesis for different eye movement measures. Signif. codes: 0.001 *** 0.05 ** *

Conclusions

- The findings on this experiment provide some evidence suggesting the VCD index's effects on saccade targeting mechanism and oculomotor control in reading natural traditional Chinese sentences.
- The PVL of First Fixation (PVL FF) and Single Fixation (PVL SF) cases was further into the word for the RB condition in comparison to the NB and LB conditions, thus evidencing that information about the luminance patterns of words located in parafoveal vision is used to some extent when specifying the next saccade target. The influence of the words' VCD on saccade targeting mechanism might take place only after readers have isolated the currently fixated word from the string of characters, which implies the awareness about the left boundary position of word $n + 1$. Thus, when LB and NB words were located in parafoveal vision a saccade strategy based on the awareness about the location of a semantic boundary is more likely to be deployed, resulting in comparable PVL for those words. Instead, the VCD index of RB words was perceived for the readers as an additional cue about how the visual components are distributed along the area of the word to be fixated next, which shifts the PVL rightwards.
- The LMMs results demonstrated the relatively large effect of launch site on the saccade targeting mechanism as indicated by the models on PVL FF and PVL SF; in consequence, when the launch site of the incoming saccade was closer to the word's beginning, the eyes were sent further into the word and the probability of making multiple fixation within that word was also reduced. Furthermore, closer launch sites allow a larger preprocessing of words located in parafoveal vision as reflected by it's significant effects on the GD and skipping probability models.
- The analyses on the duration measures indicate that readers of Chinese require more time to integrate visual and semantic features of NB words in contrast to LB and RB words, as suggested by the early processing measures (i.e. FFD and SFD) regardless of having controlled both character and word frequency, as well as the overall stroke count across conditions. In addition, the integration of visual and semantic features for the NB words was to some extent delayed as suggested by relatively larger GD times.

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